Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation

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Abstract

Automated research and self-improvement pipelines choose a reward model or LLM judge by its accuracy on a validation set. However, the pipeline or its designers often decide which items that set contains, so whoever curates the set can also choose the evaluator. We introduce *curation certificates*, which state how much of the validation set could be removed before a different model wins. For benchmarks scored by weighted accuracy, the certificate in benchmark weight is at least the score gap between the top two models, and the certificate in items is computed exactly by sorting per-item score differences. On RewardBench (118 models, 2,985 items) and RewardBench 2 (188 models, 1,753 items in its five accuracy subsets), linear programs over item weights, checked on real item subsets, show that the weight bound is nearly tight for the top models. RewardBench's winner stays first when any 11 items are removed, but removing 12 particular items puts the runner-up first, and at a curation budget of 1% of its weight any of 3 models could be the winner. RewardBench 2's winner stays first when any 42 items are removed. On random 200-item validation sets, selections whose top-two gap exceeds 2 points match the full-benchmark winner 87% of the time, against 43% for the rest. These results suggest that an evaluator selection should be reported with its certificate, and we give a protocol for doing so.

1 Introduction

Systems that automate research or improve themselves need an evaluator: a reward model to optimize against, or an LLM judge that ranks candidate outputs (Lu et al., 2024; Zheng et al., 2023). Evaluators are chosen by meta-evaluation: each candidate is scored by how often it agrees with human preference labels on a benchmark such as RewardBench (Lambert et al., 2025; Malik et al., 2026), and the top scorer is selected. However, in an automated pipeline the pipeline itself often picks the items used for this choice, as a filtered subset, a domain slice or a small validation set. If the ranking of evaluators depends on which items are kept, then whoever picks the items also picks the evaluator, which is reward hacking one level above the reward (Gao et al., 2023). This raises the question we study: **how much of the validation set must be removed before a different evaluator wins?**

Leaderboard rankings are known to shift with the choice of benchmark, the answer format and private testing (Dehghani et al., 2021; Alzahrani et al., 2024; Singh et al., 2025; Zhang and Hardt, 2024), and with targeted edits to individual items (Dubrovsky et al., 2025). Gordienko et al. (2026) count how many *datasets* a developer must train on to top a multi-task leaderboard, and show that this problem is NP-hard under mean win rate. Oyarhoseini et al. (2026) use influence approximations to show that editing under 1% of pairwise matches can change the leader of a Bradley–Terry arena. Both papers study how a ranking can be attacked. For training data, related certificates exist: dropping a small fraction of observations can overturn a statistical conclusion (Broderick et al., 2020), and certified defenses bound the damage that poisoned training points can do (Steinhardt et al., 2017). We give the analogous certificate for the evaluation data used to select a model, one that whoever selects an evaluator can compute and report, for the weighted-accuracy scores used to meta-evaluate reward models and judges. Figure 1 previews the answer: for most models, the share that must be removed to make the model win is close to its score gap to the winner. Our contributions are:

- Two curation certificates for the winner of a weighted-accuracy benchmark, a lower bound in benchmark weight and an exact

count in items, both computed from the per-item score table alone (§3).

- Evidence from linear programs that the weight bound is nearly tight: for the top 20 RewardBench models, the cheapest curation we found removes at most 1.02 times the gap (§4).
- Two uses (§4). At a 1% curation budget, 3 RewardBench models could be the winner. On random 200-item validation sets, certified selections are correct 87% of the time, against 43% for the rest.

2 Scores, Curations and Data

A benchmark has items $i = 1, \dots, N$ with official weights $w_i^0 \geq 0$, $\sum_i w_i^0 = 1$. Model m has per-item score $r_{mi} \in [0, 1]$ and official score $S_m = \sum_i w_i^0 r_{mi}$. A *curation* keeps a subset K of the items, with kept share $f = \sum_{i \in K} w_i^0$, and rescores every model on K alone: $S_m^K = \sum_{i \in K} w_i^0 r_{mi} / f$. We call $1 - f$ the *removed share*, and report scores, gaps and shares in percentage points. K *crowns* m if m scores strictly highest on it, $S_m^K > S_k^K$ for every $k \neq m$. The winner is *certified* against a class of curations if it stays first under every curation in the class.

Data. RewardBench v1 publishes per-item results for 118 models on 2,985 preference pairs in 23 subsets (Lambert et al., 2025). We rebuild each official score from them: a section score is the count-weighted mean of its subsets' accuracies, with `math-prm` upweighted as in the released code, and the overall score is the mean of the four section scores. For four models, from the winner down to a 0.5B chat model, our subset accuracies match RewardBench's published values exactly. The result files list items by position, and a check of the item text found no misaligned items. RewardBench 2 (Malik et al., 2026) publishes results by item id. We use the 188 models with results on all 1,753 items of its five accuracy subsets, score each model by the mean of its subset accuracies, and exclude the Ties subset, which uses a different metric.

3 Certificates in Weight and in Items

Let a be the winner. For each challenger c, let $g_c = S_a - S_c > 0$ be its gap to the winner and $d_i^c = w_i^0 (r_{ai} - r_{ci})$ the contribution of item i to that gap.

Proposition 1. *Any curation K on which c ties or beats a ($S_c^K \geq S_a^K$) removes a share of at least g_c. Hence the winner is certified against every curation that removes a share below $g^\star = \min_c g_c$, its gap to the runner-up.*

Proof. $f\,(S_a^K - S_c^K) = \sum_{i \in K} d_i^c = g_c - \sum_{i \notin K} d_i^c \geq g_c - (1 - f)$, since $d_i^c \leq w_i^0$; this is positive whenever $1 - f < g_c$. □

Proposition 2. *Sort the contributions so that $d_{(1)}^c \geq d_{(2)}^c \geq \cdots$, and let $k_c = \min\{k : \sum_{j \leq k} d_{(j)}^c > g_c\}$, with $k_c = \infty$ if no such k exists. Removing a set R of items makes c overtake a if and only if $\sum_{i \in R} d_i^c > g_c$. So no removal of fewer than k_c items lets c overtake a, and removing the k_c items with the largest contributions does. With $k^\star = \min_c k_c$, no challenger overtakes the winner when any $k^\star - 1$ items are removed, and some $k^\star$ items let one do so, so this count is exact.*

Proof. From the identity above, $S_c^K > S_a^K$ iff $\sum_{i \notin K} d_i^c > g_c$. Any k removed items contribute at most $\sum_{j \leq k} d_{(j)}^c$. □

Both certificates need only the table of per-item scores: $g^\star$ can be read off the leaderboard, and $k^\star$ takes one sort per challenger.

Cheapest crowning curations. Prop. 1 is only a lower bound, because removing items that favor the winner over one challenger can let a third model overtake both. To see how close the bound is, we compute for each model m the largest kept share f at which some curation crowns it. For fixed f, relaxing the subset to item weights w gives a linear program:

$$\max_{w,t} \; t \;\; \text{s.t.} \;\; \sum_i w_i (r_{mi} - r_{ki}) \geq t \;\; \forall k \neq m,$$
$$0 \leq w_i \leq w_i^0 / f, \quad \sum_i w_i = 1,$$

where every subset with kept share f is a feasible w, so $t^* \leq 0$ proves that no such subset crowns m. If $t^* > 0$, we round the solution to an item subset, keeping the items whose LP weight exceeds half their upper bound or, if that subset fails, every item with positive weight, and accept it only if m is strictly first after rescoring. We scan f downward from 1 and then bisect. Every reported cost is the removed weight of such a verified subset, and so an upper bound on the cheapest crowning removal, which Prop. 1 bounds from below. A *section-preserving* variant also fixes each section's total weight, so that the curated set keeps the

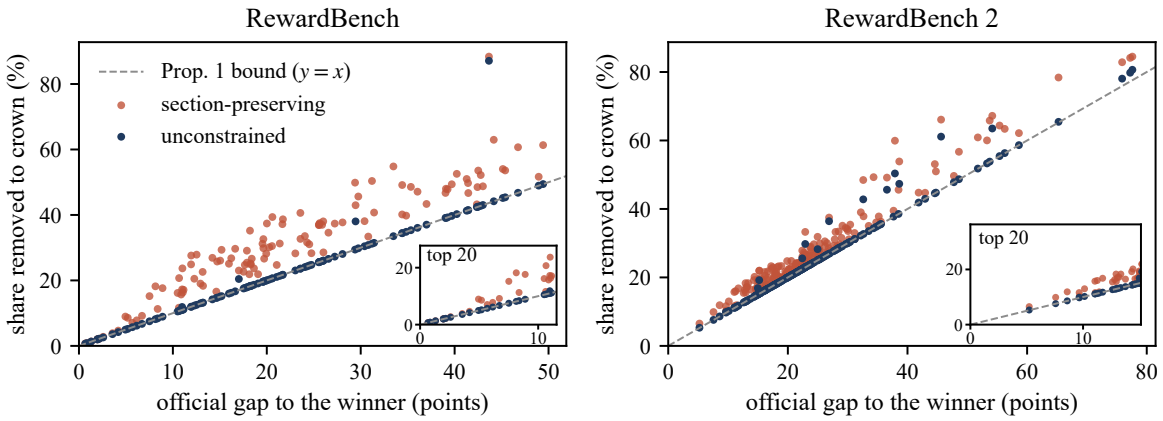


Figure 1: Share of the benchmark that must be removed to crown each model, against the model's official gap to the winner: RewardBench (118 models, left) and RewardBench 2 (188, right). Blue: unconstrained curation. Orange: curation that holds each section's share fixed. Insets: the 20 models closest to the winner. Blue points lie on the bound $y = x$ of Prop. 1 for most models, so the top-two gap is close to the true cost of changing the winner; holding sections fixed costs up to 2.31 times the gap.

benchmark's composition. Any verified section-preserving subset is also a valid unconstrained one, so we report the unconstrained cost as the smaller of the two. For ordinal aggregates such as mean win rate, the analogous selection problem is NP-hard (Gordienko et al., 2026); for weighted accuracy, each check is a linear program.

Possible-winner sets. Let the curation budget δ be the largest share that may have been removed, and let $P(\delta)$ be the set of models that some curation removing at most δ can crown. By Prop. 1, $P(\delta)$ contains only models within δ of the winner, $P(\delta) \subseteq \{m : S_a - S_m \leq \delta\}$. Every model whose verified LP cost is at most δ is in $P(\delta)$. Together these bound the size of $P(\delta)$ from both sides.

4 Results on RewardBench and RewardBench 2

RewardBench's winner falls to 12 removed items. RewardBench's winner leads by $g^\star = 0.67$ points, so it stays first under any curation that removes less than 0.67% of benchmark weight. The cheapest crowning curation we found removes 0.68%, so the certificate is tight to within 0.01 points. In items, the winner stays first when any 11 items are removed, and removing 12 particular items makes the runner-up the new winner, which we confirmed by rescoring. (The LP minimizes weight, not item count, and its curation removed 19 items.) RewardBench 2's winner leads by 5.22 points, needs 5.30% of the weight removed to lose first place, and stays first when any 42 items are removed.

Budget δ	0.5%	1%	2%	5%	10%
RewardBench	1	3	5–6	11	19
RewardBench 2	1	1	1	1	6

Table 1: Possible winners $|P(\delta)|$ at curation budget δ (% of benchmark weight). In a range a–b, a models have a verified curation that removes at most δ and b models are within δ of the winner (Prop. 1). A single number means $a = b$. RewardBench's winner is not certified at a 1% budget; RewardBench 2's is certified up to 5%.

The weight bound is nearly tight. Every model on both benchmarks can be crowned by some curation. On RewardBench, the ratio of removed share to gap has a median of 1.00 and is at most 1.02 among the top 20 models (Figure 1); the largest ratio over all models is 2.00. Every top-20 model can be crowned while keeping at least 89.5% of the benchmark. On RewardBench 2, the median ratio is 1.00 and the maximum 1.35. A curator who also keeps each section's share fixed, so that the per-section breakdown shows no trace, pays a median of 1.39 times the gap on RewardBench and 1.12 on RewardBench 2. Because the gap bounds the cheapest crowning removal from below and each reported cost bounds it from above, a ratio near 1 pins that removal down from both sides.

At a 1% budget, 3 RewardBench models could win. Because the bound is nearly tight, the two

	RewardBench			RewardBench 2		
n	all	cert.	not	all	cert.	not
100	35	64 (15)	30	87	95 (76)	61
200	46	87 (8)	43	97	100 (88)	75
500	66	98 (3)	65	100	100 (97)	97

Table 2: Selection on random validation sets of n items, drawn with replacement in proportion to official weight (2,000 draws per n). Each entry is the % of draws that select the full-benchmark winner: over all draws (all), over draws whose top-two gap on the set exceeds 2 points (cert.; % of draws in brackets), and over the remaining draws (not). Certified selections are correct far more often than the rest.

ends of $P(\delta)$ are close (Table 1). On RewardBench, a pipeline that may have curated 1% of the benchmark can only say that the best model is one of 3, and at 5% one of 11. RewardBench 2's winner is the only possible winner up to a 5% budget.

The top-two gap flags unreliable validation-set selections. In practice a pipeline scores candidates on a small validation set, and small item sets can estimate benchmark scores well (Maia Polo et al., 2024); the certificate is then the top-two gap on that set. Table 2 draws such sets at random, so that each set's score is an unbiased estimate of the official score. These draws measure sampling noise. Curation of the validation set itself is covered by Props. 1–2 applied to that set, so the same top-two gap guards a selection against both. On RewardBench, a 100-item set selects the full-benchmark winner in 35% of draws. With 200 items, the 8% of selections certified at 2 points (151 of 2,000) match it 87% of the time (Wilson 95% interval 81–92%), against 43% for the rest (41–45%). The 2-point threshold is not special: at 1 point the split is 66% against 35%, and at 5 points no selection is certified, as expected when the full-benchmark winner leads by 0.67 points. On RewardBench 2, whose winner leads by 5.22 points, most selections are certified and nearly all are correct; at 5 points the split is 100% against 94%. Using only the validation scores, the certificate separates selections that usually match the full-benchmark winner from those that often do not.

5 Protocol for Certified Selection

The certificates cover removals from a committed item set, not choices made before it was fixed, and they need per-item scores on all of it. For an automated pipeline that selects an evaluator: (i) fix the validation items and record a hash of them before scoring any candidate, score every candidate on all of them, and log every filter applied afterwards; (ii) set a curation budget δ, the share of the set that could have been dropped or filtered; (iii) report the certificates $g^\star$ and $k^\star$ with the selection, and accept the winner only if $g^\star > \delta$; (iv) otherwise, report the set of possible winners, bounded as above, instead of a single winner. On RewardBench with a 1% budget, for example, step (iv) applies: $g^\star = 0.67$ points, and the pipeline reports 3 possible winners. The certificates take one pass over the score table, so the protocol needs no extra model calls.

6 Conclusion

We introduced curation certificates for selecting a reward model or LLM judge by weighted accuracy: a lower bound in benchmark weight, equal to the top-two gap, and an exact count in items. On RewardBench and RewardBench 2 the weight bound is nearly tight for the top models, so the top-two gap is close to the true cost of changing the winner, and on small validation sets it separates reliable selections from unreliable ones. We release the code and derived data, and we hope that automated pipelines will report a certificate with every evaluator they select.

Limitations

Removal only. We study one kind of curation, removing items. A curator who could also reweight the kept items, paraphrase them or change the prompt format would have more freedom, so the certificates hold only against removal. Bounded reweighting fits the linear program of §3; paraphrasing and format changes do not.

Two benchmarks. Both benchmarks score reward models by weighted accuracy. The certificates apply to any benchmark scored this way, but not to ordinal aggregates such as mean win rate, where the analogous problem is NP-hard (Gordienko et al., 2026).

LP costs are upper bounds. The item certificate is exact. The LP costs are achieved by verified subsets but not proven minimal, so the cheapest crowning removal lies between a model's gap and its reported cost. Where the ratio is near 1 this pins

the minimum down; elsewhere the weight bound may be tighter than we report.

Random validation sets. The validation-set experiment draws random sets, not adversarial ones, so it measures sampling noise rather than curation, and it reports only thresholds of 1, 2 and 5 points.

Scope of claims. We do not claim that any published result was curated. RewardBench 2's Ties subset is excluded, so our RewardBench 2 scores differ slightly from the official ones.

Reporting and Reproducibility

All results come from public per-item result files (`allenai/reward-bench-results`, `allenai/reward-bench-2-results`). All analysis ran on one laptop CPU with SciPy's HiGHS solver: each full LP sweep takes one to two hours, and the certificates take seconds. No models were run. One failed run is worth reporting. The first section-preserving search started its bisection at a very small kept share, where rounding can empty a section, and wrongly reported that the winner could not be crowned; the corrected search scans downward from $f = 1$. Code and derived data are at `https://github.com/adrian-erlikhman/curated-to-win`.

Use of AI Assistants

An AI assistant (Claude, Anthropic) helped write the analysis code and draft the text. The author chose the study, checked the proofs and the results, and takes responsibility for the content.

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